

Anyonic braiding in a chiral Mach–Zehnder interferometer

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Fractional quantum statistics are the defining characteristic of anyons. Measuring the phase generated by an exchange of anyons is challenging, as standard interferometry set-ups—such as the Fabry–Pérot interferometer—suffer from charging effects that obscure the interference signal. Here we present the observation of anyonic interference and exchange phases in an optical-like Mach–Zehnder interferometer based on co-propagating interface modes. By avoiding backscattering and deleterious charging effects, this set-up enables pristine and robust Aharonov–Bohm interference without any phase slips. At various fractional filling factors, the observed flux periodicities agree with the fundamental fractionally charged excitations that correspond to Jain states and depend only on the bulk topological order. To probe the anyonic statistics, we used a small, charged top gate in the interferometer bulk to induce localized quasiparticles without modifying the Aharonov–Bohm phase. The added quasiparticles introduce periodic phase slips. The sign and magnitude of the observed phase slips align with the expected value at filling $1/3$, but their direction shows systematic deviations at fillings $2/5$ and $3/7$. Control over added individual quasiparticles in this design is essential for measuring the coveted non-abelian statistics in the future.

Quantum Hall states are quintessential examples of topological phases of matter¹. The fractional states host quasiparticle (QP) excitations that carry fractional charges^{2–4} and exhibit anyonic statistics^{5–8}. Upon exchanging two identical QPs, the phase of their joint wavefunction changes by a fraction of π , unlike the behaviour of fermions (bosons), where the exchange phase is π (2π)⁹. Two key methods for probing these anyonic statistics involve interference in space¹⁰ or in time^{11,12}. In the spatial domain, the phase difference between two QP trajectories that enclose a magnetic flux is the Aharonov–Bohm (AB) phase. The number of enclosed QPs determines an added statistical phase. For bosons or fermions, the statistical phase is an unobservable multiple of 2π , but it becomes measurable in systems hosting anyons^{13,14}. QPs can either be localized in the bulk (with a substantial excitation energy) or be delocalized in chiral edge modes at the periphery of the bulk (with a smaller excitation energy).

To probe these phases experimentally, interference studies of fractional quantum Hall states use two primary architectures: the Fabry–Pérot interferometer (FPI)^{10,15–19} and the Mach–Zehnder Interferometer (MZI)^{20–22}. In an FPI, a quantum Hall bulk is bound on both sides by regions with a lower filling factor ν (Fig. 1c). The interference loop consists of counter-propagating charge modes on opposite sides of the central area of the bulk. The loop is partially closed by two quantum point contacts (QPCs), which allow tunnelling only of specific charges whose quantization is set by the bulk filling. Figure 1c illustrates the essential features of an FPI for the integer case using semiclassical electron trajectories. In particular, closed paths within the central region correspond to localized QPs within the interference loop. Promoting electrons to composite fermions results in localized fractional QPs. Excitation of the latter changes the phase of the interfering QPs loops in discrete steps, also known as phase slips. In reality, transmission

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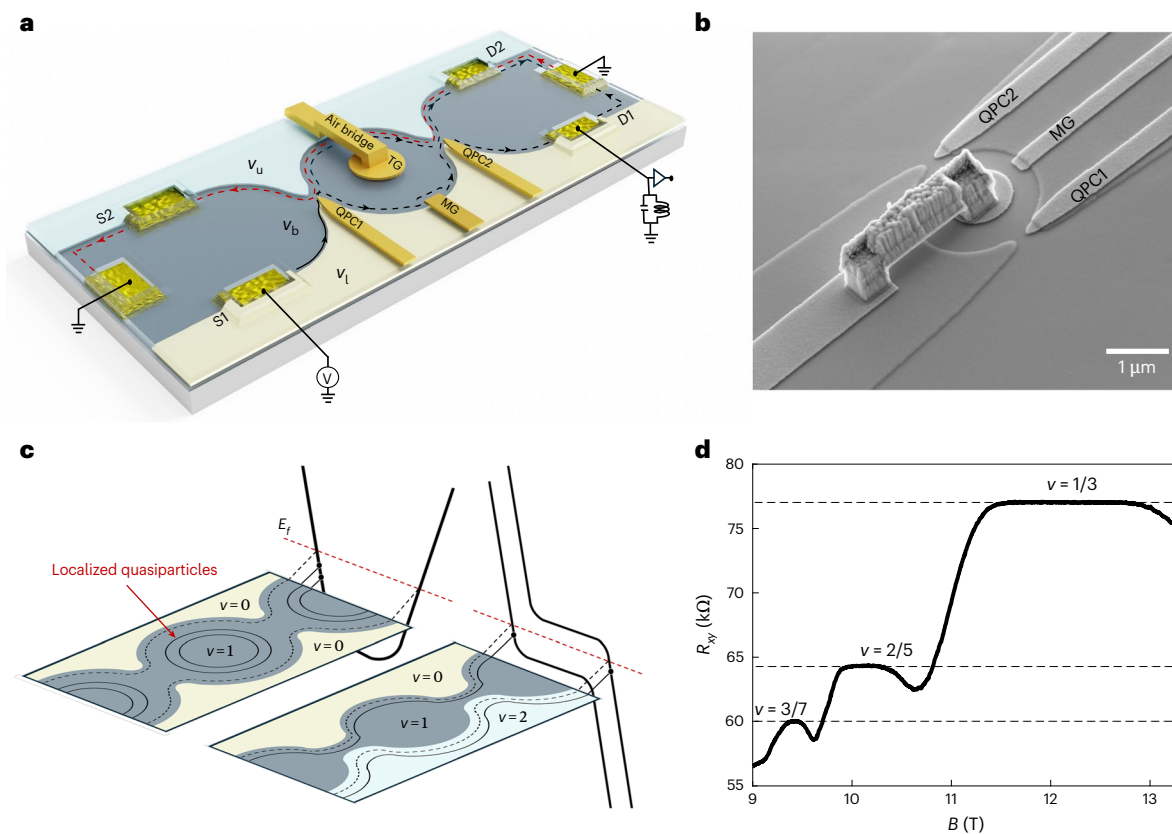


Fig. 1 | OMZI and FPI. a, Schematic of the OMZI. The lower gate (light yellow with filling factor ν_l) is biased negatively, depleting the density. The upper gate (light blue with filling factor ν_u) is biased positively, increasing the density. Co-propagating modes (downstream in black) are partitioned at QPC1 and QPC2. The source S1 is charged and the interfering signal is measured in drain D1. Upstream neutral modes are drawn as dotted red lines. An anti-dot in the middle of the OMZI bulk is formed by a negatively charged TG connected by an air bridge

to a power supply. **b**, Scanning electron micrograph of the OMZI. **c**, Comparison of an FPI and an OMZI for integer quantum Hall states based on semiclassical electron trajectories. In the FPI (left), states below the Fermi level E_F form closed orbits corresponding to localized QPs. Fully or partially closed trajectories suffer from Coulomb interactions. In the OMZI (right), all trajectories extend beyond the central region and are insensitive to charging effects. **d**, The Hall resistance (R_{xy}) is in the range of the fractional states under study.

through the FPI is additionally influenced by a breathing of the AB area with the magnetic field and, most importantly, by intrinsic charging energy for adding QPs to its central region^{23–27}. Still, sufficient screening of the Coulomb interaction allows the observation of phase slips, which approximately agree with the expected braiding phases^{26–29}.

By contrast, a conventional MZI contains an internal grounded drain, which renders it free of charging effects. However, this drain also makes it subject to the Byers–Yang argument³⁰, which predicts $1\Phi_0$ periodicity³¹. Physically, this periodicity arises because QPs returning to the internal loop continuously ‘dress’ the trivial AB interference of fractional charges by their braiding phase^{32–34}. Phase slips are neither observed³⁴ nor expected in the MZI³²; however, the visibility profile (as a function of the MZI transmission) drastically differs from that in the integer regime, consistent with its origin in anyonic interference³⁴.

An optical-like Mach–Zehnder interferometer (OMZI, read as OM-ZI) combines the immunity of an MZI to charging effects with the ability of an FPI to observe marked AB (super-)periodicities and anyonic braiding. An early version of an OMZI was first proposed by Giovannetti et al. in 2008³⁵ and tested by Deviatov et al. in 2012³⁶. A theoretical study by Batra et al.³⁷ presented an exact solution for the interference of specific edge modes³⁷. The OMZI structure is drawn schematically in Fig. 1a. It comprises three regions, each with a different filling factor. The central (bulk) region is tuned to filling ν_b and is bounded by two regions that are gate-tuned to an upper filling $\nu_u > \nu_b$ and to a lower filling $\nu_l < \nu_b$. This configuration results in two co-propagating interface charged modes with corresponding fillings: $\nu_u - \nu_b$ and $\nu_b - \nu_l$ (ref. 38). Current emanating from source S1 (or source S2) is partitioned by

the input QPC1 to two forward-propagating chiral interface modes surrounding the bulk. The two interface modes are re-partitioned by the output QPC2, and the current is eventually collected by drain D1 (or drain D2). The chirality of the charged modes ensures that all the sourced current passes through the OMZI and is not subject to charging effects. Additionally, the potential landscape, which has a saddle instead of a valley in the central region, implies the absence of localized low-energy QPs inside the interference loop (Fig. 1c). These two features distinguish an OMZI from an FPI and allow it to exhibit pristine AB interference without phase slips.

To observe anyonic braiding, the QPs must be introduced into the OMZI by a separate mechanism. For this purpose, we employed a top gate (TG) that locally partially depleted or accumulated charge upon application of a negative or positive gate voltage, respectively. A reduced filling factor under the TG led to an interface mode in its periphery, which supported fractionally quantized charged excitations (encoded as the winding number of the interface mode). These QPs had lower energies than bulk QPs and, thus, could be pushed across the Fermi level by varying the TG voltage or the magnetic field. Thereupon, fractional charge transferred from the gapless outer boundary of the OMZI to the inner edge mode encircling the small, gated region. The resulting QPs in the centre of the OMZI were spatially isolated from the interfering QPs (at the far edge) and affected them solely through the quantum statistical phases.

Our OMZI (Fig. 1a,b) was fabricated in a two-dimensional (2D) electron gas, induced in a GaAs–AlGaAs heterostructure, with a density of $1.07 \times 10^{11} \text{ cm}^{-2}$ and dark mobility $4.5 \times 10^6 \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$ at 4.2 K.

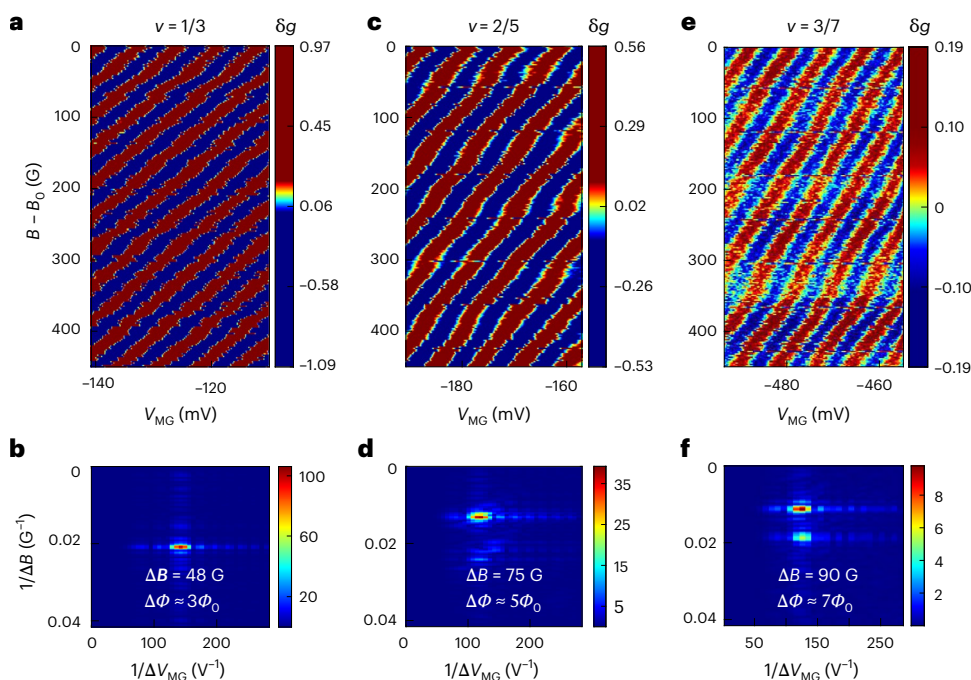


Fig. 2 | AB pajamas plots of co-propagating edge modes in an OMZI. a–f, 2D AB interference as a function of magnetic field B and the modulation-gate voltage V_{MG} . a, c, e, Pajamas corresponding to the partitioning of the innermost modes such that the interfering QPs carry charges $e/3$, $e/5$ and $e/7$. a, $\nu_b = 1/3$. c, $\nu_b = 2/5$.

e, $\nu_b = 3/7$. b, d, f, Fast Fourier transforms reflect the expected flux periodicities, $\Delta\Phi = (e/e^*)\Phi_0$. b, $\nu_b = 1/3$. d, $\nu_b = 2/5$. f, $\nu_b = 3/7$. Phase slips related to QP excitations are absent, as expected. B_0 , reference field on the plateau; δg , proportional to conductance.

Measurements were performed at an electronic temperature of approximately 15 mK. The fabricated internal area of the OMZI $A = 3 \mu\text{m}^2$, with a single path length of $3 \mu\text{m}$. Because of depletion of the gates, the actual OMZI area was smaller and affected by the voltages of the gates and of the two QPCs. The isolated TG, with fabricated area $A_{TG} = 0.785 \mu\text{m}^2$, was charged through an air bridge (Fig. 1b). A small a.c. signal (approximately $2 \mu\text{V}_{\text{RMS}}$ (micro-volt (μV)) at a frequency of 776 kHz was applied to source S1. The interfering signal reached the two drains, and the signal was monitored in D1. The signal was filtered by an LC circuit (with bandwidth $\text{BW} = 30 \text{ kHz}$) and subsequently amplified by an amplifier cooled to 4.2 K (gain = 10) and then by a room-temperature amplifier (gain = 400). A spectrum analyser, tuned to the input carrier frequency, averaged the amplified output signal over a narrow bandwidth, $\text{BW} = 5 \text{ Hz}$.

We tested three configurations with distinct fractional bulk fillings: $\nu_b = 1/3$, $2/5$ and $3/7$. The upper and lower gated regions were selected to realize two co-propagating interface modes formed in the following configurations: $\nu_u - \nu_b - \nu_l = 2/3 - 1/3 - 0$, $1 - 2/5 - 0$ and $1 - 3/7 - 0$. Upstream neutral modes at the upper positively gated interface accompanied the co-propagating interface charge modes (Fig. 1 and schematic in Supplementary Information Section 11) and were expected to suppress the visibility (indeed, the visibility was 2–5%). The two QPCs in the OMZI were tuned to partition the innermost interface mode in all three configurations, that is, the one carrying $1/3$, $1/15$ and $1/35$ (in units of the quantum conductance). The average transmission was kept at $t_{\text{SI-D1}} = 0.8$ at $\nu_b = 1/3$ and $t_{\text{SI-D1}} = 0.5$ at $\nu_b = 2/5$ or $3/7$, all with TG voltage $V_{TG} = 0$. The flux periodicities in the three configurations matched the expectations based on the partitioned fractional charges at the two QPCs (Fig. 2). The actual inner area of the OMZI was not known accurately (being depleted near the $\nu = 0$ side and expanded near the accumulated side), yet it was also dependent on ν_b . It was probably smaller than the fabricated area (by approximately 5–10%). For $\nu_b = 1/3$, the observed periodicity matched the expected value of approximately $3\Phi_0$, with the actual OMZI area taken to be $2.6 \mu\text{m}^2$. Similarly, at $\nu_b = 2/5$ and $3/7$, we found values of approximately $5\Phi_0$ (actual area $2.76 \mu\text{m}^2$) and approximately $7\Phi_0$ (actual area $3.2 \mu\text{m}^2$).

The periodicities in V_{MG} obeyed $B \times \Delta V_{MG}^i = 85 \pm 3 \text{ T mV}$. As anticipated, for $V_{TG} = 0$, the $B - V_{MG}$ pajamas were free of phase flips. Note that interference of integer modes, tested in configurations 2–1–0, 3–2–0 and 4–2–0 (Supplementary Information Section 5) was unsuccessful due to a lack of partitioning in the QPCs, probably because of the opposite spins of the co-propagating integer edge modes.

The visibility of the interference signal is defined as $v_e = (t_{\text{max}} - t_{\text{min}}) / (t_{\text{max}} + t_{\text{min}})$, where t_{max} and t_{min} are the maximum and minimum transmissions at each drain (peak and valley of the AB oscillations), respectively. The observed visibilities varied around 2–5% at an average transmission $t_{\text{SI-D1}} = 0.8$ for $\nu_b = 1/3$ and $t_{\text{SI-D1}} = 0.5$ for $\nu_b = 2/5$ and $3/7$ (inner modes at higher fillings). The visibilities of the outer modes of $\nu_b = 2/5$ and $3/7$ were comparable to those of the inner modes. We attribute the substantially smaller visibilities in the OMZI compared to a conventional MZI²⁰ to the topological neutral modes at the interface (Supplementary Information Section 10).

The theoretically expected braiding phases for Jain states have been obtained in several studies^{7,9,13,27,39,40}. These incompressible fractional quantum Hall states arise at fractional fillings, $\nu = n/(2pn + 1)$, with the integer n denoting the number of filled composite-fermion levels and p the generation of composite fermions. Here, $p = 1$ corresponds to two-flux quanta attached to the electrons. Elementary QPs of the Jain states carry a fractional charge, $e^* = e/(2pn + 1)$ and obey fractional statistics. Exchanging two QPs yields a phase factor $\pm 2\pi p/(2np + 1)$, depending on the orientation of the exchange path. The two interfering trajectories in the OMZI differ by a full loop around the interferometer centre and are expected to acquire a quantized phase difference of $\theta^*/2\pi = 2p/(2np + 1)$ for each localized QP³⁹. Here, we consider the fillings $1/3$, $2/5$ and $3/7$, with the expected braiding phases $\theta^*/2\pi = 2/3$, $2/5$ and $2/7$ (Supplementary Information Section 2).

To observe these braiding phases, we studied the AB pajamas at a range of TG voltages, $V_{TG} = -(50\text{--}100) \text{ mV}$, for the three bulk filling configurations with the two QPCs tuned to partition the innermost edge mode in each of the three configurations. The quantized phase slips (Fig. 3) were observed approximately when the magnetic flux piercing

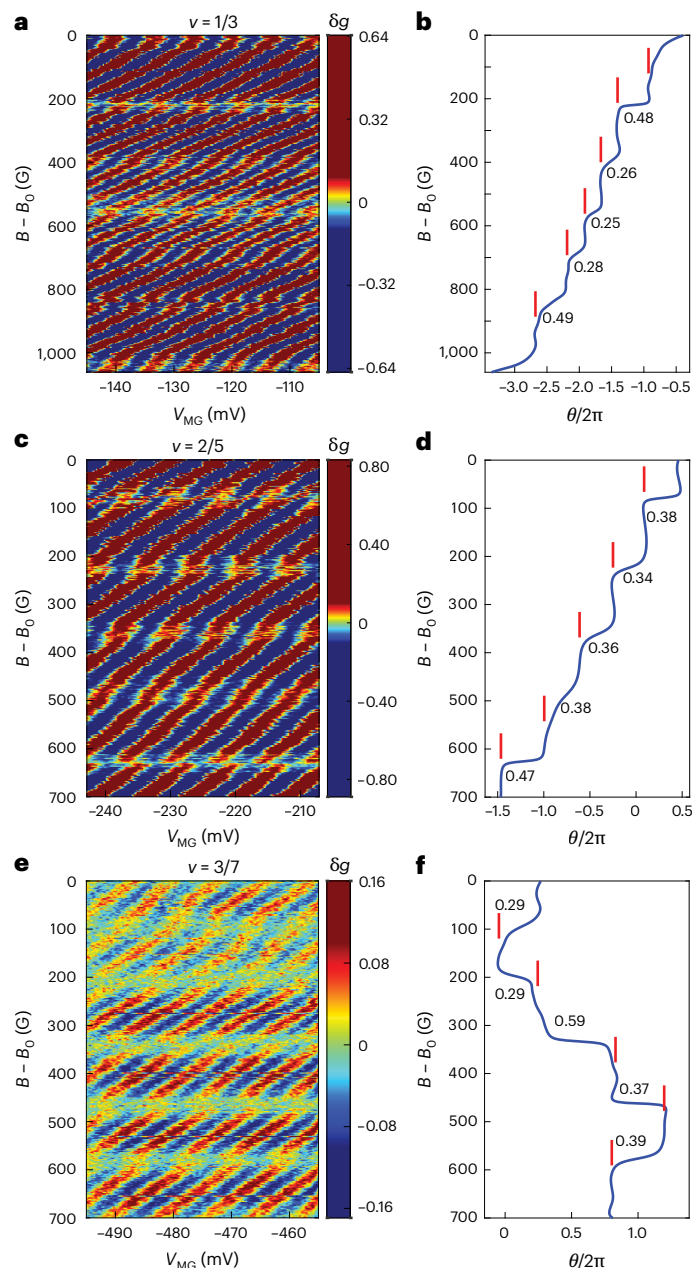


Fig. 3 | AB pajamas with magnetic field excitations of local QPs. **a, c, e**, 2D AB interference as a function of magnetic field B and the modulation-gate voltage V_{MG} . Clear phase slips appear for approximately every flux quantum piercing the ‘effective area’ under the TG. The TG of the anti-dot is charged with a voltage in the range -50 to -100 mV, leading to weak depletion under the gate, **a**, At $\nu_b = 1/3$, $V_{TG} = -90$ mV. **c**, At $\nu_b = 2/5$, $V_{TG} = -50$ mV. **e**, At $\nu_b = 3/7$, $V_{TG} = -70$ mV. The fabricated area of the TG was $0.78 \mu\text{m}^2$; however, the effective area extracted from the distance between phase jumps was $0.27 \mu\text{m}^2$ at $\nu_b = 1/3$, $0.3 \mu\text{m}^2$ at $\nu_b = 2/5$ and $0.34 \mu\text{m}^2$ at $\nu_b = 3/7$, indicating a smaller depleted island under the TG. **b, d, f**, Phase slips calculated with the lock-in technique (Methods and Supplementary Information Section 3). The vertical red lines indicate the phase in regions where it is constant. The phase shifts between two consecutive regions are labelled. The magnitudes of the phase slips (modulo 2π) generally agree with the theoretically expected value $\theta^*/2\pi = 2p/(1 + 2np)$ (Supplementary Table 1). **b**, At $\nu_b = 1/3$, all phase jumps are consistent with the expected creation of quasi-holes as the magnetic field was increased. **d**, The jumps at $\nu_b = 2/5$ instead correspond to inserted QPs. **f**, At $\nu_b = 3/7$, both types of phase jumps are present.

the partly depleted area under the TG changed by $1\Phi_0$ (independent of the modulation-gate (MG) voltage, V_{MG}). The phase-slip magnitudes determined either directly from the 2D pajama plots or with the lock-in procedure (Methods and Supplementary Information Section 3) generally agree with the (absolute) expected phase slips (Supplementary Table 1). Studies of the interfering partitioned outer modes in the QPCs and the corresponding phase slips are presented in Supplementary Information Section II. Note that the signs of some phase slips at filling $\nu = 2/5$ and $3/7$ are opposite to the theoretical expectations. Increasing the magnetic field reduced the filling factor, which was expected to create quasi-holes inside the interferometer. Instead, the observed phase jumps at $\nu = 2/5$ corresponded to added QPs. At the filling factor $\nu = 3/7$, jumps were observed in both directions. A possible explanation for the unexpected phase-slip signs could arise from the energetics under the TG, which may favour the simultaneous introduction of more than one quasi-hole (or QP). For example, four $e/5$ quasi-holes at $\nu = 2/5$ are equivalent to one integer hole and one $e/5$ QP. The interference phase was sensitive only to the fractional charge and would, thus, exhibit phase slip associated with one QP added under the TG.

The most striking observation of anyonic braiding in the OMZI occurred when B and V_{MG} were held constant, which allowed the AB phase to remain unchanged while V_{TG} was varied. This process introduced QPs into the centre of the bulk without disturbing the interference pattern, enabling precise detection of phase slips. This test was performed in the configurations $\nu_b = 1/3$ and $2/5$. A 2D plot of the OMZI conductance as a function of V_{TG} and B is shown in Fig. 4 (being a different pajama). At a constant magnetic field near a conductance step, a minute voltage change applied to the TG ($\Delta V_{TG} \approx 1$ mV) led to a sharp jump in the conductance of the OMZI (Fig. 4). To translate the conductance steps to a phase slip in the AB interference, Fig. 4b,d shows the AB oscillations (as a function of B on both sides of vertical dashed lines that mark the conductance steps). The abrupt shift in the AB oscillation (across $\Delta V_{TG} = 1$ mV) directly reveals the quantized phase slip due to added QPs at the TG. The sizes of the phase slips are provided in the caption of Fig. 4.

Our experiments used three different modes of the OMZI: (1) Without charging the TG, we found pristine AB oscillations with flux periodicities tied to the fundamental charge of the QP. (2) With a charged TG ($V_{TG} \neq 0$), localized QPs were introduced periodically with a changing magnetic field. Consequently, the OMZI showed phase slips corresponding to the braiding phase of the QPs in the AB interference pattern. (3) At a constant magnetic field and a constant MG voltage (V_{MG}), localized bulk QPs were introduced one by one with V_{TG} , leading to abrupt slips in the AB phase, as was expected due to QP braiding.

Similar tests performed with different co-propagating interface modes proved that bulk filling—and not the interface modes—determines the interfering charges and their exchange phase. The versatility of the OMZI geometry requires minor modifications to realize different probes of QP statistics. For example, time braiding^{11,12} can be performed using the first QPC to dilute the QP current, which will braid with thermally excited QPs in the second QPC. Attaching two OMZIs will form a Hanbury, Brown and Twiss configuration^{22,35}, allowing intensity interference of QPs arriving from two different sources. In addition, it will be easy to perform Hong, Ou and Mandel experiments on QPs⁴¹. Moreover, the OMZI is an ideal interferometer for obtaining interference and determining the topological order of non-abelian fractional quantum Hall states.

Online content

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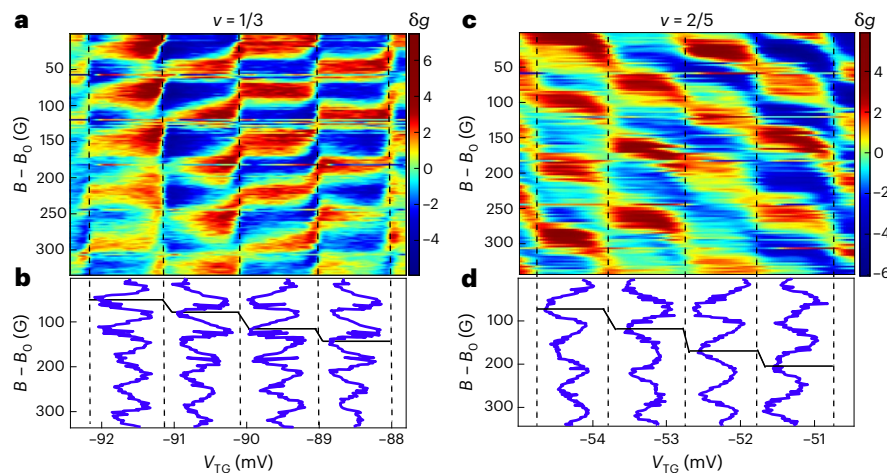


Fig. 4 | Local excitation of QPs by activating V_{TG} at the local anti-dot TG. **a,c**, 2D plots of the AB interference for $\nu_b = 1/3$ (**a**) and $2/5$ (**c**) of partitioned innermost edge modes as a function of the magnetic field B and the TG voltage V_{TG} . At any constant B , changing the applied voltage V_{TG} by $\Delta V_{TG} = 1$ mV led to sharp phase jumps indicated by the vertical dotted lines. **b,d**, Conductance

oscillations with B for each region separated by phase jumps. Comparing two neighbouring regions yields the actual phase difference. **b**, For $\nu_b = 1/3$, the average magnitude of the jumps was $\theta^*/2\pi = 0.38$. **d**, For $\nu_b = 2/5$, it was $\theta^*/2\pi = 0.42$. At $\nu_b = 3/7$, the visibility was too low to extract reliable values (see expected value for θ^* in Supplementary Table 1).

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Methods

Sample fabrication process

The mesa of the device, which had dimensions $250 \times 650 \mu\text{m}^2$, was prepared in GaAs–AlGaAs heterostructure by wet etching in $\text{H}_2\text{O}_2:\text{H}_3\text{PO}_4:\text{H}_2\text{O} = 1:1:50$ for 100 s. The depth of the 2D electron gas was 170 nm from the surface. The ohmic contacts were deposited at the edge of the mesa as well as within the mesa (for the interface modes) in the sequence Ni (15 nm), Au (260 nm), Ge (130 nm), Ni (87.5 nm) and Au (15 nm) from the top GaAs surface. This process was followed by annealing at 440°C for 80 s. The sample was covered by 30 nm of an HfO_2 layer to electrically separate the ohmic contacts from the large metal gates (Ti (5 nm)/Au (15 nm)). Two large metal gates separated the mesa into three regions with filling factors ν_{L} , ν_{B} and ν_{R} . In the following step, the mesa was covered again by a 25-nm HfO_2 layer, followed by the deposition of QPCs, the MG and the TG. In the final step, ohmic contacts, large metallic gates, QPCs and the MG were connected to pads with thick gold lines. The TG was connected by an air bridge in the interferometer region, followed by gold lines that passed over the HfO_2 that coated the metallic gates.

Geometric phase analysis

The phase slips were analysed using the spatial lock-in method. Namely, the AB oscillations described by $T(\mathbf{r}) = A(\mathbf{r}) \sin(\mathbf{q}\mathbf{r} + \theta(\mathbf{r}))$, where $\mathbf{r} = (B, V)$, $\mathbf{q} = (q_B, q_V)$ is the frequency of the AB oscillations and $\theta(\mathbf{r})$ is the phase shift due to the phase slips. To extract $\theta(\mathbf{r})$, two lock-in signals were defined: $s_1(\mathbf{r}) = \cos(\mathbf{q}\mathbf{r})$ and $s_2(\mathbf{r}) = \sin(\mathbf{q}\mathbf{r})$. The original signal $T(\mathbf{r})$ was multiplied by these signals, and low-pass filtering was applied to obtain outputs proportional to $\sin \theta(\mathbf{r})$ and $\cos \theta(\mathbf{r})$. The ratio of these allowed us to calculate the phase slips as

$$\theta(\mathbf{r}) = \arctan \frac{\text{LPF}(s_1(\mathbf{r}) \times T(\mathbf{r}))}{\text{LPF}(s_2(\mathbf{r}) \times T(\mathbf{r}))}$$

(Supplementary Information Section 3).

Data availability

Source data are provided with this paper. These data are also publicly available via Zenodo at <https://doi.org/10.5281/zenodo.15394075> (ref. 42).

References

42. Ghosh, B. Anyonic braiding in a chiral Mach-Zehnder interferometer. *Zenodo* <https://doi.org/10.5281/zenodo.15394075> (2025).

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Author contributions

B.G. fabricated the devices. B.G. and M.L. performed the measurements and analysed the data with input from M.H. L.M. characterized the devices at the initial stage of the experiment. M.H. supervised the experimental design, execution and data analysis. D.F.M. worked on the theoretical aspects and data analysis. V.U. grew the GaAs heterostructures. All authors contributed to the writing of the paper.

Competing interests

The authors declare no competing interests.

Additional information

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